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Collaborative Mobility-Aware Resource Allocation Framework Using Evolutionary Lion Optimization for Next-Generation Mobile Networks

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ABSTRACT: In terms of computer networking, Software-Defined Networking (SDN) provides a fast, convenient, maneuverable, and cost-effective method for controlling network flow with a minimum investment to reduce costs and increase benefits for massive mobile users. Future cellular networks face new challenges given the explosive growth of mobile data traffic and the limited availability of spectrum resources. Particularly, resource management in terms of load sensing and handover decisions in heterogeneous networks has become critical. Recently, the Group Mobility Model (GMM) named MoMo was introduced to solve the handover decision issue. The technology which distributes traffic among neighbouring Access Points (APs) to reduce congestion over an area is called Resource management and through which the optimal use of resources can be achieved by ensuring that wireless traffic has a proportionate share. To provide seamless and ubiquitous wireless connectivity to mobile users, efficient mobility management is necessary to ensure Quality of Service (QoS). In this study, an SDN-based Group Mobility Model (GMM) and resource management algorithm by Mutation Lion Optimization Algorithm (MLOA) is proposed to alleviate handover and address issues arising out of congestion in the network. The proposed method researches on the SDN controller in a global perspective of the given network for achieving network conditions and the end user QoS requirements into consideration. The proposed method shall be used dynamically for managing the flow of traffic and to support a large number of mobile users in a given SDN with optimum resource utilization. SDN-GMM provides transparent and efficient mobility support for ongoing sessions in each handoff scenario, thus eliminating congestion overhead and packet loss related to mobile traffic, and as a result, improving QoS for mobile users. The performance analysis through experiments depicts that the proposed model gives enhanced performance when compared to that of the other conventional methods and also has proven efficiency in terms of other parameters such as latency, cost of signaling, throughput and packet losses. The results from the simulation show that the proposed method, GMM-MLOA, greatly improves the performance of handovers and maintains optimum resource utilization and efficiency.

KEYWORDS: Group Mobility Management (GMM), Software Defined Network (SDN), Quality of Service (QoS), Future Mobile Networks, Mutation Lion Optimization Algorithm (MLOA), MoMo model, handover.

1 Introduction

The rapidly growing mobile traffic in recent years is expected to reach 77.5 EB per month by 2022 and account for 71% of total internet traffic. The growth in mobile traffic is accelerating owing



to the advent of smartphones and the expansion of video services. Most of the network service providers face more issues, such as heavy traffic and optimum resource allocation for complicated services, which require more connectivity requirements and thus the need for maintaining a seamless service amid diverse technologies and changing business requirements has become a mandate [1-2]. The network operators are hence in need of an innovative solution for performance improvement and efficiency, and also to minimize the operational cost and maintenance. In this connection, 3GPP, the third-generation partnership project, have proposed a flat optimization method such as the local IP and selective IP for traffic offload and IP flow mobility (IFOM) [3]. In order to combat the limitations of current centralized mobility management systems, the IETF created the Distributed Mobility Management (DMM) Working Group, which specifies an IP mobility management architecture separating the data plane from the control plane [4-5]. Design and development of flat architectures are being designed for improving the data plane, but still the cost was not under control, and also the complexity of controlling the plane was increasing in orders [6]. SDN combined with Network Functions Virtualizations (NFV) helps in decoupling the data and planes to offer an altogether new architecture of network, which is tailor-made for improvement in managing the control plane at both the backhaul as well as core network level [7]. In this way, the network operator shall reduce the overall operational cost and maintenance for the deployment of the service.

Particularly, the separation of the data transfer and control parts of a network device through SDN can be successfully integrated with DMM technology in accordance with the method of distributing mobility functions in DMM. DMM schemes to enhance network structure by dispersing mobility functions have been researched. The DMM scheme can prevent the problem of excessive traffic concentration on some specific devices (e.g., Home Agent (HA) and Local Mobility Agent (LMA)) and improve the reliability of the network by solving the problem of service interruptions in the case of device failure. Additionally, by moving the mobile anchor closer to the user, the packet delivery path can be optimized, and effective interworking can be ensured with a content delivery network. However, the disadvantage of the DMM scheme is largely overcome by reducing the burden of signaling processing by designating a specific router as a centralized mobile anchor for some sessions.

Group Mobility Model (GMM) is a mobility pattern described to predict the movement of the mobile node in terms of continuous changeover time, such as velocity, location, and acceleration [8]. In this model, each of the mobile nodes moves randomly together within the group using the Random Waypoint approach [9] and Reference Point Group Mobility Pattern (RPGP) [10]. Both group mobility and individual mobility are used DynnaMo approach, which provides a memory-related model. The MoMo model combines a memory-based individual model and a flexible group model, which increases the accuracy of the mobility model [11]. In order to combat the limitations of current centralized mobility management systems, the IETF created the Distributed Mobility Management (DMM) Working Group, which specifies an IP mobility management architecture separating the data plane from the control plane. Due to the fact that existing protocols were designed at present levels of network density, they will become a bottleneck in dense small cell deployments, as well as being used more frequently to communicate mobile management messages. OF-SDNs are yet another candidate solution which largely supports DMM features and exceeds the performance of conventional solutions. There are numerous advantages of OpenFlow, which make it more familiar in the use of SDN-based solutions. This includes reaching out to the plane, which is being forwarded, that switches between the network and services. In this research paper, SDN SDN-based MoMo model is proposed for future mobile networks to reduce the handover latency, signaling cost, and packet drop ratio. OpenFlow-Based Group Mobility Model (GMM) with an integrated DS for mobile services in SDNs. The contribution of the proposed SDN-based GMM work comprises of following steps: The proposed model, the Mobility Management Module (MME) and the Admission Control module are

incorporated into the SDN architecture. The Mutation Lion Optimization Algorithm (MLOA) is proposed to alleviate handover and solve the congestion issues. The proposed method addresses the SDN controller view in a global perspective and the end user QoS requirements addicted to consideration. In the handover phase, the grouping of each node is checked periodically based on two conditions, namely, the Free State and Forced State processes. Finally, the performances such as latency, throughput, cost of signaling and losses are then assessed for the proposed work, which shows an improved efficiency than the existing GMM technique.

2 Literature Review

Duan et al [12] introduced SDN-enabled resource optimization techniques applied in futuristic cellular networks. In the proposed method, three objectives were considered, namely the shortage of spectrum through effective offloading of traffic, addressing of congestion in the network through optimal load balancing across different cells and to achieve these while maintaining the expected QoS. In this method, SDN enables partial offloading of data, and load balance techniques are proposed. The technique identifies the opportunities in a global perspective for achieving optimum resource allocation. The proposed method is analyzed for its performance in a simulated environment. A framework is also proposed for the quantification of delays which are incurred by virtue of SDN-enabled processing of data. Simulation results have shown that the proposed method of load balancing considerably increases the extent of equilibrium and the stability of the network when compared to that of baseline methods. The proposed method also shows a satisfactory result in QoS while also achieving significant savings in the usage of cellular resources.

Yang and Kim [13] applied the SDN concept to the Distributed Mobility Management (DMM) architecture to support efficient mobility management and to solve existing problems in centralized mobility schemes. In our proposed scheme, we define a new mobility management entity for reducing tunnelling overhead, supporting route optimization. To evaluate the proposed management scheme, establish an analytical model and formulate signaling and packet-delivery costs. Nguyen et al [14] proposed a DMM-enabled solution, which is based on the architecture of SDN, named S-DMM. The proposed technique gives more advantages that including a deployment-free mobility component which accesses the router and is independent of the underlying methods and flow-based mobile support. The experimental results have proven that the S-DMM architecture is more stable and also scalable when compared to conventional DMM. The results have also proven that the proposed S-DMM has zero penalty on performance (in terms of latency and delay) when compared to that of the conventional DMM, yet the cost is a bit higher in management, which makes the proposed technique best suitable for the upcoming 5G network.

Ko et al [15] introduced an SDN-based technique for DMM, which is different from the normal network and terminal-based techniques. The proposed SDN-DMM is implemented on the location and handover management in the centralized controller when the packet forwarding functions are completely distributed in routers. This makes the SDN-DMM achieve packet forwarding optimizations and also provides constant benefits, such as networks and to manage traffic effectively.

Jeon et al [16] presented the Distributed Mobility Management (DMM) as a core idea for tackling the hindrances which can impact present networks by presenting the deployment of various points of anchors which are much closer to the terminus. There were various proposals for building DMM-based solutions that focus on the merits of futuristic networks that aim to overcome limitations in the present centralized MMS. A detailed analysis of the experimental results is carried out, and the essential design constraints and options are also identified through the impact analysis on performance. To add this, the recent advancements in DMM with the emerging techniques, such as control and separation of the data plane, are also left for future study, which can impact the network paradigm for the forthcoming generation.

Khan et al [17] analyzed the effect of various system variables on overall handover delays in SDNized wireless networks and designed diverse mobility management approaches. The authors have also studied various enhancements in handling delays caused during handovers in a couple of aspects. (i) when the proactive MMA algorithm is implemented, and (ii) when the delegation of partial control of MMA is implemented on the controller. To implement the proposed techniques in an OF network, the authors also suggest different potential extensions for the OpenFlow-based protocol. The experiments are performed in a large-scale demonstrator that exhibits proactive results which are better than the reactive measures for the topologies that are smaller in size. Further, the authors have suggested a Long-term Evolution (LTE) that has specific delegation control as a case study.

In Park et al [18], SDN have been proposed to provide mobility management based on OpenFlow (OMM) and a data structure for mobility services. Mobility Management Entities (MMEs) are incorporated into one scheme to manage mobility functions and to organize a new data structure based on Local Mobility Anchors (LMAs) and Mobile Access Gateways (MAGs). MAG and LMA entries that have duplicate content are eliminated, and the entries that have duplicate content are integrated.

Cominardi et al [19] introduced the proxy-based Mobile IPV6 solution using the DMM, where it serves as a definite baseline which explores DMM standards in SDN, which also include the detection of DMM, its principles and challenges. Based on the investigation, the SDN-enabled DMM is evaluated with the standard methods in both analytical and experimental views.

A Bursty Multi-node Handover scheme (BMH-DMM) architecture was used in Huang et al. [20] to deal with this problem of mobile internet in part by utilizing partially distributed mobility management. For a group that has n MNs, the proposed BMH-DMM method eliminates n sets of control messages for handover from a group of them with one set of control messages. The analysis shows that the amount of handover overhead does not increase so much when the number of MNs switch over at the same time and place, which means the proposed BMH-DMM method can reduce the bandwidth consumption for exchanging handover signaling messages by about 30%. Additionally, using the proposed BMH-DMM scheme is less likely to cause network utilization, handover delay, packet loss, and blocking during the handover process than using the individual handover model.

Bi et al [21] introduced an SDN modelled Mobility Manager scheme to provide support for the seamless domain handovers along with optimized routes. By decoupling SDN mobility management from packet forwarding, SDN-MM provides route optimization and mobility control logic in an SDN controller while excluding traffic redirection from it. An extensive set of signaling operations is created by means of SDN-MM in order to provide transparent and efficient mobility support for ongoing sessions in a handover scenario, which mitigates packet loss and unnecessary tunnelling overhead, and accordingly improves QoS for mobile users. In SDN-MM, an SDN controller calculates the most efficient end-to-end route in advance of handover and decides whether to migrate traffic to the route based on the performance gain and the signaling overhead, which greatly increases bandwidth utilization. The simulation results show that the proposed SDN-MM enhances handover performance and ensures high resource utilization efficiency at the same time.

In order to evaluate the accuracy of this Make-Before-Break (MBB) method, Jaumard et al [22] developed a near-optimal two-phase defragmentation process to assess its accuracy. To calculate the best rerouting, e.g., with the least bandwidth requirements, first construct an exact optimization model for the decomposition scheme. Second, demonstrate the best MBB reroute option if possible, with the resultant re-routing; otherwise, demonstrate no option. Numerous experiments in many datasets are performed in small and medium-sized datasets to compare the proposed method with that of the two-phased models. Finally, the focus was made on the real-time scalability of the proposed method in large and real-time data.

3 Proposed System

In this study, an SDN-based Group Mobility Model (GMM) and a resource management algorithm by Mutation Lion Optimization Algorithm (MLOA) are proposed to alleviate handover and to address the network congestion issues. Group-based Mobility Management (GMM) method, named MoMo, is proposed by utilizing Distributed Mobility Management (DMM) is introduced to SDN environment. By using an SDN controller's global view of the network, the proposed algorithm takes network conditions and end-user QoS requirements into account to achieve network quality. An SDN controller's global view of the network and the optimization of resource allocation decisions are handled by MLOA. The proposed system has four phases of operation, namely the Initial Phase, Handover (HO) Preparation phase, Resource management phase, and HO Phase.

3.1 Network Model

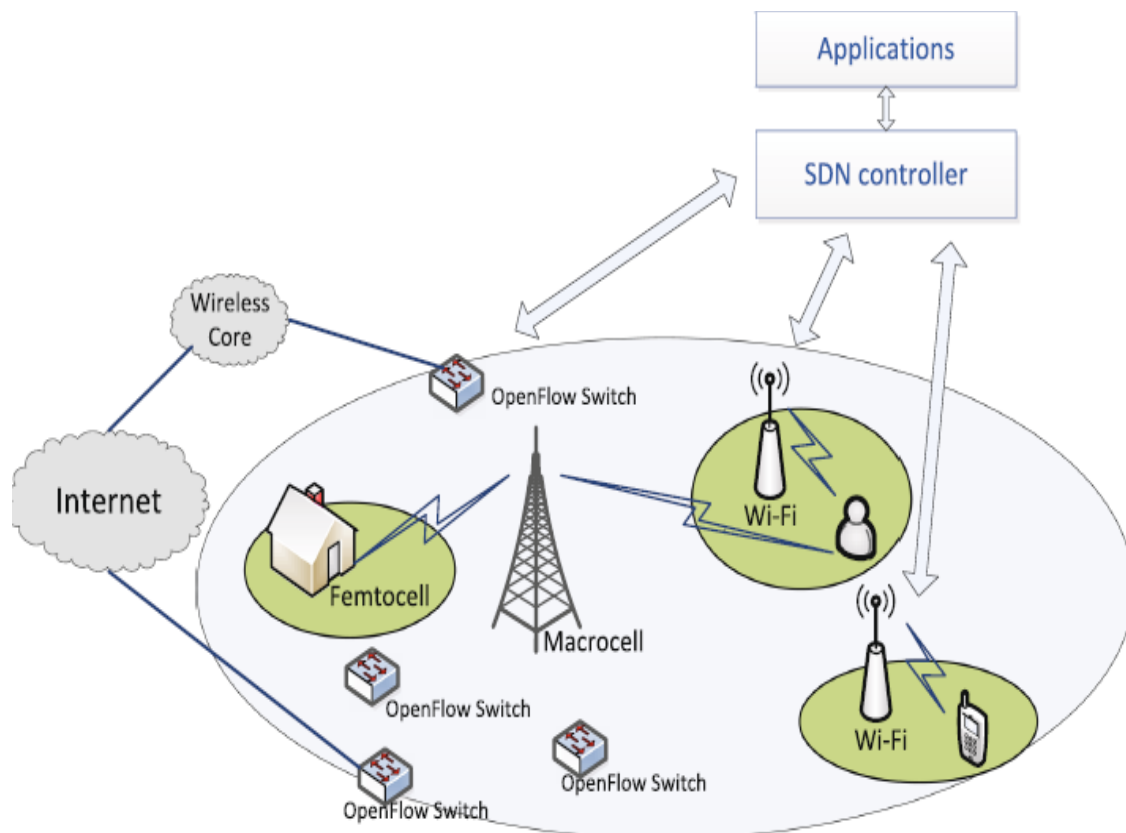


Figure 1: SDN-based wireless heterogeneous network structure with control plane design

Consider Figure 1, where the environment of a network with cellular BSs and Wi-Fi access points (APs) is shown. A licensed communication is used by BSs, while an unlicensed communication is used by Access points, which are present in a random manner inside each cell. OpenFlow-based protocol is followed in BSs, APs and switches for enabling ease of control to the SDN controller using a secured channel. Assumption is made in such a way that every BS and the access points are strongly connected with the OpenFlow switches as presented in Figure 1. For a given cellular network, these switches are then co-located with the Base stations. Different from macro cell switches, the femtocell directly connects to the internet, whereas macro cells are then a part of core networks. On the other

side, as far as the access points are considered, these switches will be found located inside the ISP and every switch is then connected to many access points. All the switches will then be found controlled in a centralized manner using SDN, and given the incidence that the proposed controller is just a program which runs on the server, the same can be fixed anywhere inside the network, even in a distant data center [23].

In Figure 1, the OpenFlow switch is similar to an access point, except that its original control functions have been transferred to the controller, leaving it with only data forwarding functionality, which makes it equivalent to a pure data channel. Controlling devices, managing OpenFlow switches and APs, and delivering flow tables is the function of the controller, which is a centralized and unified logic control unit. Furthermore, in this architecture, the SDN controller is a fundamental component, and as a fundamental component of the architecture, the SDN framework must be outlined and discussed in depth. The controller application consists of two core modules, namely, Offload Manager (OM) and Resource Management (RM). As part of the OM module, the service-level rules describe the traffic features and characteristics that are necessary for the offload manager, such as the corresponding IP address, port number, bit rate, and delay sensitivity, based on the TFT [22] and the corresponding QoS descriptions. Additionally, the RM module features load measurement and mobile management features, which execute algorithms to measure and balance cell load. The overall framework of the proposed method is presented in Figure 2, where it is observed that SDN applications can make use of the global view of access points to improve the management of a network.

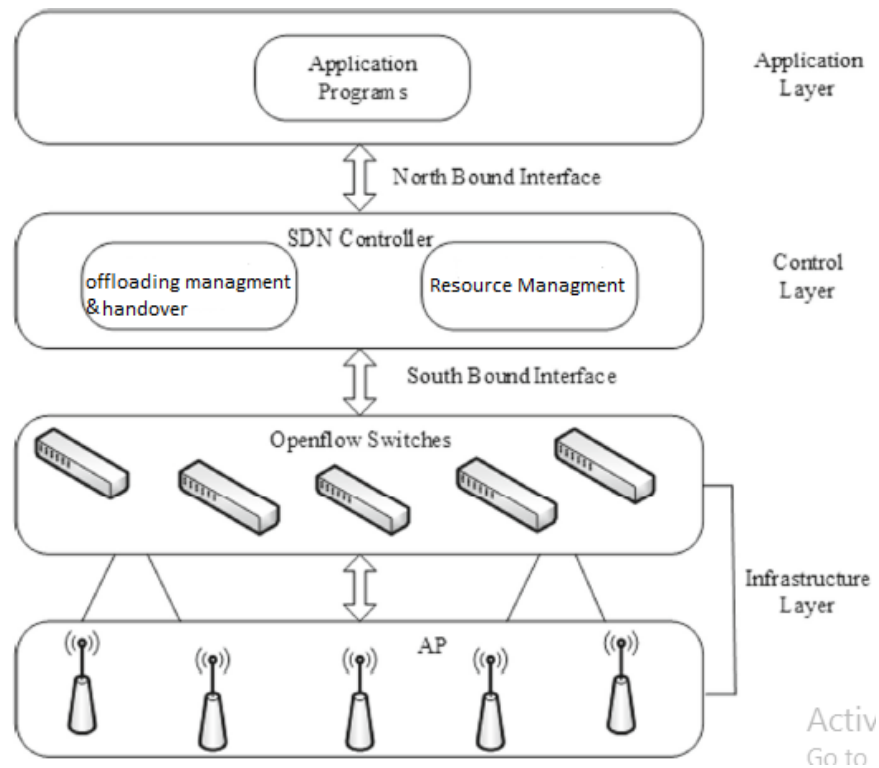


Figure 2: SDN-based offloading scenario and module framework

The focus of this section is to present SDN-based resource management algorithms for the heterogeneous network shown in Figure 2. The aim is to solve spectrum shortage problems and network congestion in cellular networks by enabling mobile data offloading and Resource Management within the SDN. An SDN-based model in which partial offloading of data carried out is

depicted in the model. As the process indicates, here, only a part of the data is transferred to the Wi-Fi and the rest of the traffic is then transferred across various cellular networks. For further elaboration, the network will be unable to satisfy the requirement due to delay and loss-sensitive flow, as only part of the data is offloaded. It is shown in the latter part of this paper how the proposed method can help in improving the performance by considering various requirements. The access points in a network are then stripped of the management of control functions, and each of the access points can be connected with different mobile terminals. There exists a module for managing a larger volume of access points with the terminals in the network, and hence, a module for management is deployed in the controller for managing the terminals. The proposed model is segregated into sublevels: The access point management and the terminal management. The access point management helps in discovering and configuring the APs, and in managing and collecting the information in each of the access points.

3.2 Offloading Management

The proposed algorithm proceeds as follows: The data is queued in the OpenFlow switch and waits for further processing; OM selects traffic flows based on delay thresholds when a mobile user moves within the Wi-Fi coverage, where optimum traffic selection is made for offloading. Given the fact that these Wi-Fi networks give high rates in the absence of QoS, which guarantees the unlicensed bands, the privilege to offload traffic shall be made discrete. The aim is to minimize the usage of cellular leverage using Wi-Fi connection as available while performing the same in the absence of any performance degradation. The algorithm used for the appropriate flow is presented below.

Step 1: The OM collects the requirements of traffic flow from all users inside the Wi-Fi area, and the resources required are calculated. It is assumed that every user has a certain demand for traffic represented by u_i and a rate of link r_i^w for the access points. Each user of the Wi-Fi has a resource demand, calculated as $\theta_i^w = u_i/r_i^w$. The resources of Wi-Fi that are available for the offloading shall be estimated based on the same.

Step 2: If Wi-Fi cannot transfer all the data before the deadline, partial offloading is performed. The OM calculates the amount of data that can be transferred over the Wi-Fi network within the delay tolerance threshold, and in other words, all the data will be offloaded to the Wi-Fi network. The delay in tolerance of the threshold is calculated (Th_s) and the volume of data represented by V_s which are transferred in a given time frame Th_s . With the effect of parallel transmission, the bursts are subjected to arrive at the receiver's end in a similar time and thus which minimises the period of wait that is required for every packet to reach the receiver's device. Hence, the overall time delay for the arrival of the packet is minimized using the offloading method.

Step 3: The Om then updates the records of the Wi-Fi sources available after the process of offloading and QoS, which can be provided during the PDO and these records are kept for a while, and it helps in quick decision making where a new flow of traffic arrives. The direction of the mobility of a particular user is estimated by OM and assuming that a user enters and then leaves an offloading region at two different times, namely ti_{in} and ti_{out} and as a consequence, the time of connection would be $ti_c = ti_{out} - ti_{in}$. Here, the tolerance of the delay threshold Th_s will be equal to that of the difference among the applications in a specific deadline, represented as Th_d , which are then obtained through TFT filters of the network model and hence the waiting time in the SDN process is represented as D, which is explained in the following section. Let the amount of data transmitted be represented as V_s . Let bw_1, bw_2 represent the bandwidths that are allotted by the cellular network and the Wi-Fi network, respectively. As represented in [24], the smallest value among the ti_r and Th_s shall be used for the calculation of the present volume of data. Hence, V_s is denoted as,

$$V_s = bw_1 Th_s + bw_2 \min(ti_r, Th_s) \quad (1)$$

3.3 Resource management

It consists of two modules: the resource sensing and the handover decision module, deployed at the application layer. The management module installed in the controller is responsible for obtaining most of the state information about each of the access points, and the load sensing module is mainly responsible for obtaining the state information about the network. As a consequence of the AP weight algorithm proposed in this paper, each AP is assigned a weight, and each terminal is switched according to the weights. This balances network resource management by scheduling traffic and switching terminals appropriately. An improved seamless handover technique and a method for resource management handover are proposed, which can improve QoS and resource utilization of networks.

3.4 Seamless handover

The terminal handover delay and the network delay both occur in four steps: queuing, node processing, transmission, and propagation, with delays in the first three steps being on the order of a few microseconds and hence the proposed technique mainly concentrates on the delays due to handover and propagation. The traditional handover method usually takes hundreds of milliseconds [25]. The handover phase comprises two modules, such as the MMM and the admission controlling module and the dummy and tiny cells are administered in the data planes for managing mobility. Furthermore, the Mobility Management Entity (MME) is made to handle the HO procedure. In this way, the required mobility-related data, for instance, the MN subscription data, mobile identification, and tackling trail area posts are attained. The mobile node ID (MN_ID) and Evolving Node ID (EN_ID) parameters are employed for MNs and small cells [25]. In the proposed SDN-GMM, MoMo model, the number of evolving nodes in the network contains a hexagonal architecture. This utilizes the Automatic Neighbour Relation (ANR) function of evolving node count by which the neighbour relation tables are updated. Accordingly, the proposed SDN-GMM MoMo model reaches the valid neighbour relations of the evolving new nodes from these tables. In future mobile networks, the random-access channel (RACH) is utilized by MN to start the session with an arbitrary access preamble throughout the primary movement of the connection process. In addition, this preamble comprises the MN_ID. Next, the goal evolving node manages the OF table to discover the arriving MN_ID. If this MN_ID is integrated into the table, an HO response is given to the MN. This response specifies that the HO appeal is established by the evolving node. Then the connect process continues amid the MN and evolving nodes with the equivalent message series as Downlink Shared Channel (DL-SCH), uplink shared channel (UL-SCH).

We investigate delays observed in the HO training stage, and if no MN_ID was found, we transfer the request to the controller, where a revision of the set of OF tables is made, and we proceed to the HO phase, which describes how the nodes of the SD, where an intelligent and futuristic network lies.

3.5 Distance among nodes

The proposed SDN-GMM-based MoMo model defines the binding conditions related to physical proximity between the nodes. The binding condition between two nodes, namely i and j are in the same group, are referred to as group mates, and is defined as in equation (2),

$$d_{ij} \leq D_c \quad (2)$$

Where, $d_{ij} = d_{ji}$ is the distance between nodes i and j . If the binding condition in equation (2) is satisfied, the two nodes are said to be distance D_c connected. Consider a group of size M . For the generic node j th set of M_j^c group mates that the node detects as connected are called its connected set. The ratio between M_j^c and the total number M of group mates is called the grouping factor ρ_j in equation (3),

$$\rho_j = \frac{M_j^c}{M-1} \quad (3)$$

The behaviour of node j depends on the Grouping Condition (GC) defined on ρ_j in equation (4),

$$\rho_j \geq \rho_{\min} \quad (4)$$

where the grouping threshold $\rho_{\min}$ is a tunable model parameter. Every node occasionally verifies whether the GC is satisfied, with period ∇u based on the result, the node transitions into one of the two subsequent states, like free and forced: Free State occurs when the GC is satisfied. Here, the node freely moves based on the boundless mobility model; a forced state occurs when the GC is not satisfied. Here, the node travels towards the closest group mate, k , is not part of its connected set, to improve its grouping factor. The speed variables v and θ are set as in equations (5) and (6),

$$v = v_{\max} \quad (5)$$

$$\theta = \begin{cases} \min(\beta_{kj}, \theta_{\text{old}} + \gamma_{\min} T_{lu}), & \text{if } \beta_{kj} \geq \theta_{\text{old}} \\ \max(\beta_{kj}, \theta_{\text{old}} - \gamma_{\max} T_{lu}), & \text{otherwise} \end{cases} \quad (6)$$

Where, $\beta_{kj} = \arctan\left(\frac{y_k - y_j}{x_k - x_j}\right)$, (x_k, y_k) and (x_j, y_j) are the locations of nodes k and j , respectively, θ_{old} is the previous direction, and T_{lu} is the time elapsed until the final position update. Equation (5) and Equation (6) ensure that the node j attains the chosen group mate k in the shortest probable time frame, while evading although destruction of the limitation on linear and angular speed. The MN is still associated with the previous Access Points (AP) that detect the next AP. It established a signal strength which is advanced than the predefined value of threshold and which is the uppermost on all of the APs that are in range. The MNS throws back reports are messages where the next information of the access point is conveyed to that of the first one.

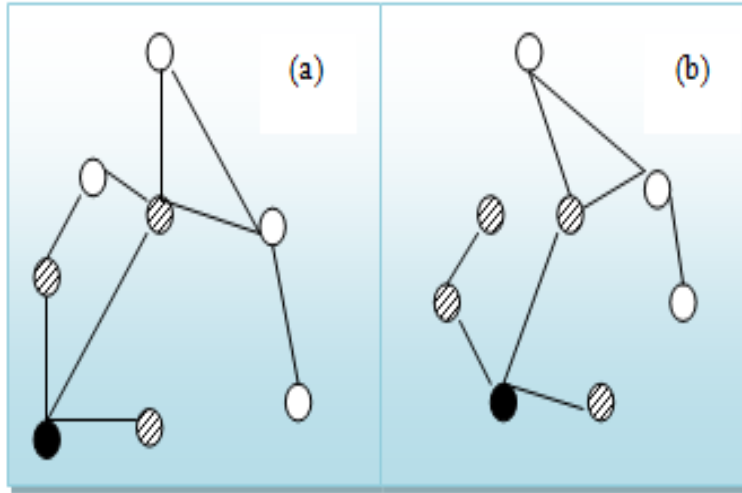


Figure 3: Example of the function of the SDN – GMM Momo mobility model for a group of 8 nodes

Figure 3 a), the measured node (black node) is linked to three striped nodes in its collection, out of its 7 group mates, and the grouping factor $\rho = 3/7$ calculated by the node is less than the necessary $\rho_{\min}$. As a result, the node goes into forced mode and travels towards the nearby group mate among its

connected set. The node preserves this behaviour until the GC is satisfied, evolving towards the situation in 2. b), where the range of the connected set is increased to 4, corresponding to $\rho \geq \rho_{\min}$. The behaviour defined in the forced mode, which is moving towards the closest group mate not in the connected set, is not the only possible one. More complex behaviours, e.g., moving towards the centroid of the points of group mates, can be easily established in the framework of the proposed SDN-GMM MoMo model. Figure 3 demonstrates an instance of the application of the SDN-GMM MoMo model in the case of a group of 8 nodes with $\rho_{\min} = 0.5$. Lines between nodes indicate connectivity for the SDN-GMM MoMo mobility model. In Figure 3a), the size 3 of the connected set for the black node leads to a grouping factor $\rho = 0.43$. The GC is thus not satisfied, and the node moves to the closest group mate not part of its connected set, until the condition is satisfied, as in the configuration shown in Figure 3b, in which $M_c = 4$, $\rho = 0.57$. The definition of connectivity and the corresponding meaning of the threshold D_c is a key feature in SDN-GMM MoMo. The replication for a flexible definition of the theory of association, rooted in the application scenario:

Connectivity related to radio communications - here, two nodes are associated either during a straight radio connection (physical layer associativity), while they are in the radio range, or during conveying, assured by additional group mates (network associativity);

Connectivity is rooted in a radio-independent constraint - for instance, if a collection will become a security team, material visibility may correspond to association: a team member will go freely until it is capable of viewing a minimum count of team members, and travels nearer to the extra members of the team when the circumstance is not met any longer.

3.6 Propagation delay

In the future, the flow table in the AP will still be updated when the terminal performs the handover, but the terminal associated with the AP has changed. Through the management modules, all the underlying APs and all the mobile terminal's state information are collected, including the parameters affecting AP performance, and are passed to the load balancing module. The load balancing module determines the merits of each AP's comprehensive performance by calculating its weight value according to the state information collected by the algorithm, before deciding whether to hand over to that AP which performs better. By using this handover method, you can balance the load on the network and improve its efficiency. Assume that the signal strength S , then S is an average of the signal strength S by equation (7).

$$S = \alpha \bar{S}_t + (1 - \alpha) \bar{S}_{t-1} \quad (7)$$

Where α is a smoothing factor. In this work, the AP historical traffic of the latest n unit time ti_0 of the AP is counted. The unit flow of the AP in the i^{th} ti_0 time by equation (8),

$$T(i) = \frac{F_i^{Tx} + F_i^{Rx}}{ti_0} \quad (8)$$

where F_i^{Tx} represent that the traffic which is sent by the access points in the i^{th} unit under a specific time ti_0 of a queue where F_i^{Rx} represent traffic that is received in the access points of the i^{th} unit in a time ti_0 of the queue. The throughput $\bar{T}$ of the access point is presented in Eq. (9),

$$\bar{T} = \frac{\sum_{i=1}^n T(i)}{n} \quad (9)$$

Based on the known number of APs and the historical performance of the AP, propose a resource index based on the throughput and total number of AP terminals. The index of the resource is represented as AP_k on a given network is R_k which is again computed using Eq (10),

$$R_k = \frac{\tilde{T}_k}{T_{\max}} + \frac{N_k}{N_{\max}} \quad (10)$$

where $\tilde{T}_k$ denote the old throughput in AP_k for a given period of time. $T_{\max}$ represent the maximum amount of throughput which the AP_k shall withstand. N_k denote the total terminals which are associated with the AP_k . $T_{\max}$ and $N_{\max}$ are obtained through intense testing. Hence, it can be concluded that these performance weights for an AP are then calculated through the Eq (11),

$$w = \frac{S_k}{L_k} \quad (11)$$

To sum up is calculated by equation (12),

$$w = \frac{\alpha \bar{S}_t + (1-\alpha) \bar{S}_{t-1}}{\frac{\tilde{T}_k}{T_{\max}} + \frac{N_k}{N_{\max}}} \quad (12)$$

In the resource balancing handover method, the controller monitors the AP through its global perspective and allocates appropriate weights for each AP based on the collected AP state information. In order to serve an STA, the controller picks the AP with the best performance according to the weight size of the AP associated with a strong signal, a relatively small number of STAs, and a low historical throughput.

3.6.1 Parameters affecting AP performance

As a general rule, the signal strength S of APs and the number of connected terminals N affect their performance, and the stronger the received signal strength, the better the wireless link's transmission quality. Handover based on signal strength has been traditional. The number of terminals associated with each AP can provide some indication of the AP load, and in general, terminals associated with the same content are selected. The potential of every terminal to contest for a channel is constant. If there is a large number of access points associated with a given terminal and if the competition is mode, the impact will also be great in the resultant collision in performance.

3.7 Mutation Lion Optimisation Algorithm (MLOA)

This is a population-based method of developing a weighted solution in SDN [26,27] wherein the first step is to create a random population over the solution space, referred to as a "Lion" weight value. In given N_{var} dimension optimisation problems, a Lion is denoted as $Lion = [x_1, \dots, x_{N_{var}}]$. N -pop solutions are generated randomly in the search space for delay comparison and throughput estimation of SDN networks. The percentile of N generated values of weight and traffic values are selected in nomad lion representation based on equations (13 and 8) as factors to decide delay. The remaining population are randomly segregated into P no of prides. Each of the solutions present in this technique has a specific gender and remains constant throughout the optimisation, and to eliminate the fact, in every pride, a percentage of S in the overall population is obtained in the last step, which are known as females, and the rest are treated as males. In the process of searching, the lion makes its best visit position.

3.7.1 Hunting

A few females hunt for prey (weight calculations, delay calculations, and throughput calculations) and in a group in order to provide food to their pride, and these females do not use the same strategies to catch the prey in every pride. A dummy sort of prey denoted as PREY is then considered at the centre and represented as hunters ($PREY = \sum hunters(x_1, \dots, x_{N_{var}})$). During the process of hunting, the hunters are often selected sequentially for the calculation of delay and for computing throughput in SDNs, and each of the hunters who are selected on a dummy prey for which the procedure is defined at a later

part, as per the group which has selected the lion (where the weight of the access points and SDN) belongs. During the whole process of hunting, in case a particular hunter moves its fitness through eq (12), the PREY shall escape from the hunter, the fresh PREY is obtained through equation (13),

$$PREY' = PREY + rand(0,1) \times Per_i \times (PREY - Hunter) \quad (13)$$

Where the PREY is in the present position of the prey for the calculation of the weight of an access point with any weight by altering the path to the nearest access point. The hunter will be in a new position, who can now attack the prey, and the PI denote the percentile of the improvement in the level of fitness of the hunter through the reduction in traffic and which increases the throughput of the access points. The changed positions of the hunters belonging to both left as well as right wings are then generated as in Eq. (14).

$$Hunter' = \begin{cases} rand((2 \times PREY - Hunter), PREY), & (2 \times PREY - Hunter) < PREY \\ rand(PREY(2 \times PREY - Hunter)), & (2 \times PREY - Hunter) > PREY \end{cases} \quad (14)$$

Here, PREY represent the present position of the selection of an alternate access point along with the values of weight in SDN. Hunter denotes the present position, and the new position is represented as Hunter'. The central hunter is generated using the Eq (15),

$$Hunter' = \begin{cases} rand(Hunter, PREY), & Hunter < PREY \\ rand(PREY, Hunter), & Hunter > PREY \end{cases} \quad (15)$$

In both of the above equations, the function $rand(a, b)$ is used to generate a number between the interval $[a, b]$, where a and b denote the upper and subsequent lower bounds. In the present work, we are introducing the mutation operator for the generation of a and b . The mutation value adds new a' and b' values via the upper and lower bounds based on the throughput speed of the terminals in the AP. It may increase the weight value of the AP and reduce the traffic congestion of the AP in the network.

3.7.2 Moving Toward a Safe Place

As discussed in the previous section, in every pride, a few females go hunting. The females which remain go towards one of the territories. As the territory related to every pride has a personal best (AP) of each of the members, which assists in Mutation Lion Optimization (MLO) for saving the better solutions obtained till time in due course of iteration and can be used as valuable and reliable input for solutions to improve in MLOA. Hence, the new position of a female shall be obtained through the Eq (16).

$$\begin{aligned} \text{Female Lion}' &= \text{Female Lion} + 2D \times rand(0,1)\{R1\} + U(-1,1) \times \tan(\theta) \times D \times \{R2\} \\ \{R1\}. \{R2\} &= 0, ||\{R2\}|| = 1 \end{aligned} \quad (16)$$

In the above equation, Female Lion indicates the present position of the female lion, and D denotes the distance in-between the position and a selected point, which is chosen by the tournament and computed using equation (2). $\{R1\}$ denote a vector that starts in a prior location of a female lion, i.e the access points (AP), and it is directed towards the given AP's position. It is seen that $\{R2\}$ will be perpendicular to that of $\{R1\}$. Initially, the success of a lion is defined if it improves its own personal best in the last iteration of MLOA. Given a group P , the success of a lion 'i' (AP) at a given iteration 't' is then defined as in Eq(17).

$$S(i, t, P) = \begin{cases} 1 & \text{Best}_{i,p}^t < \text{Best}_{i,p}^{t-1} \\ 0 & \text{Best}_{i,p}^t = \text{Best}_{i,p}^{t-1} \end{cases} \quad (17)$$

where $\text{Best}_{i,p}^t$ indicate the best position of AP identified by a lion 'i' in an iteration 't'. A large number of successes ensure that these lions have converged on an access point that has minimized traffic, which

is far away from the point of optimization. By using the success values of $K_j(s)$ shall be computed using Eq(18).

$$K_j(s) = \sum_{i=1}^n S(i, t, P) \quad j = 1, \dots, P \quad (18)$$

In the above equation, n denotes the total count of lions in a given pride (AP), $K_j(s)$ denote the total count of lions in a given pride j that has an improvement in the fitness from the previous iterations. This enables the size of the tournament to be adaptive for every iteration. It indicates that as the success rate decreases, the size of the tournament will increase, which can lead to diversity. Hence, the size of the tournament shall be calculated using Eq. (19).

$$T_j^{\text{Size}} = \max\left(2, \text{ceil}\left(\frac{K_j(s)}{2}\right)\right), j = 1, 2, \dots, P \quad (19)$$

3.7.3 Moving Toward a Safe Place

As part of the Mutation Lion Optimization Algorithm (MLOA), roaming is a strong local search and results in the Lion moving in the direction of the selected territory by x units, where x is a random number with uniform distribution.

$$x \sim U(0, 2 \times d) \quad (20)$$

In this equation, d represents the distance between the male lion's location and the selection of territory, and angle $*$ represents the angle added to this direction to determine throughput history when searching for a wider area AP. In equation (21), we calculate the new AP position of nomad lions using the MLOA algorithm based on the nomad lions' adaptive roaming assist to avoid local optima.

$$\text{Lion}'_{ij} = \begin{cases} \text{Lion}_{ij} & \text{if } \text{rand}_j > \text{pr}_i \\ \text{RAND}_j & \text{else} \end{cases} \quad (21)$$

Here, Lion_i denote the present position of AP of the i^{th} lion, j denotes the dimension, and RAND_j denotes a uniform number allotted randomly in the range $[0, 1]$, RAND indicate a randomly generated vector in every search space, and pr_i indicate the probabilities which are computed for every lion which is a nomad and are independently calculated using Eq(22).

$$\text{pr}_i = 0.1 + \min\left(0.5, \frac{(\text{Nomad}_i - \text{Best}_{\text{nomad}})}{\text{Best}_{\text{nomad}}}\right) \quad i = 1, 2, \dots, \text{number of nomad lions} \quad (22)$$

Nomad_i and $\text{Best}_{\text{nomad}}$ depict the computational cost of the present position of the i^{th} lion (AP) present in nomads, along with the corresponding best cost.

3.7.4 Mating

According to the equations below, following the selection of the female and male (s) for mating, the new cubs are created. Mating is an essential process, as it involves a linear combination between parents (APs).

$$\text{Offspring}_j1 = \beta \times \text{Female Lion}_j + \sum_{i=1}^{\text{NR}} \frac{(1-\beta)}{S_i} \times \text{Male Lion}_j^i \times S_i \quad (23)$$

$$\text{Offspring}_j2 = (1 - \beta) \times \text{Female Lion}_j + \sum_{i=1}^{\text{NR}} \frac{\beta}{S_i} \times \text{Male Lion}_j^i \times S_i \quad (24)$$

This formula is as follows: $j = \text{dimension}$, $S_i = 1$, otherwise it equals 0, $NR = \text{number of males in the pride}$, β is a randomly generated number from a normal distribution with a mean value of 0.5 and a standard deviation of 0:1, where any one of the offspring is randomly selected as a male and the other as female.

3.7.5 Migration

For the migration operator, some females (AP with reduced traffic) were selected randomly and became nomads. When selected females (APs with reduced traffic) are migrated from a pride and become nomads, the new and old females are arranged according to the weight of the access points (APs). The best females are then selected from the group in a random way and are distributed to the prides to fill the empty place of a migrated female (AP with minimized traffic). This methodology helps to maintain the entire population and shares information between the prides.

4 Results and Discussions

In this section, the HO presentation is evaluated with the proposed GMM- MLOA model and estimates performance metrics including signaling cost, average HO latency, and packet loss rate. The simulation result of the proposed method is estimated by using the NS-3 network simulator version 3.26. It is a discrete-event network simulator using the OpenFlow module. By using it, the HO process is verified. The simulation environment comprises four sorts of network essentials with 50 MUs, 10 routers that manage 10 APs, 10 corresponding nodes, and 802.11n infrastructure. The simulation is continuously performed 30 times to discover the regular outcomes. In order to evaluate the proposed method of handover, the performance indicators that are more intuitive and which reflect the performance of the network, such as cost of signaling, Latency in handover, throughput, packet loss, rate of transmission and delay, are considered.

4.1 Signalling cost

It is defined as the HO update of mobility binding acquired during the processing of Hos. Therefore, the mobile management protocol will cost more when signaling messages are delivered via wired and wireless networks, depending on the number of hops, the dimension of the message, and the weighting scale, respectively. The proposed SDN-GMM, the global assessment makes provision for the SDN controller to gain agents' routers and for managing the signal messages in processing of MN's HO. Both the MME and the MODULE are expected to hold the mobility administration and to handle MN's registration to have GMM service. The signaling cost for the proposed SDN-GMM is represented as in Eq. (25).

$$\text{cost} = \mu \left[\begin{array}{l} hL_{\text{inform}}\text{hop}_{MN-\text{router}} \\ + \rho L_{hi}\text{hop}_{\text{router}-\text{router}} \\ + \rho L_{\text{hack}}\text{hop}_{\text{router}-\text{router}} \end{array} \right] \quad (25)$$

Where h and ρ are wired, and wireless link's weight factor, μ is the arrival rate of each MN via router, L_{inform} refers to the dimension of handover message, L_{hi} and L_{hack} indicate the dimension of HI and Hack message respectively, hop specifies the connection between two nodes. Figures 4 and 5 portray the difference in the signaling cost based on the diverse velocity of MNs and with varying hop counts. It is seen that the velocity of MN is increased from 5 to 30 m/s, and when the hop count is set between 1 and 10, the radius of the corresponding binding message of the proposed GMM-MLOA is smaller when compared to conventional approaches such as CMM [28], S-hDMM [29], GMM [30], and SDN-GMM. The proposed technique achieves 187, 243, 300, 357, 419, and 498 signaling costs per packet versus mn's velocity from 5 to 30 m/s. Similarly, considering the signaling cost versus hop count and it shows that the proposed method attains 390 signaling cost per packet from 10 hop count,

whereas other methods, such as CMM, S-hDMM, GMM, and SDN-GMM, have a higher signal cost of 807, 784, 677, and 439, respectively, per packet from 10 hop count.

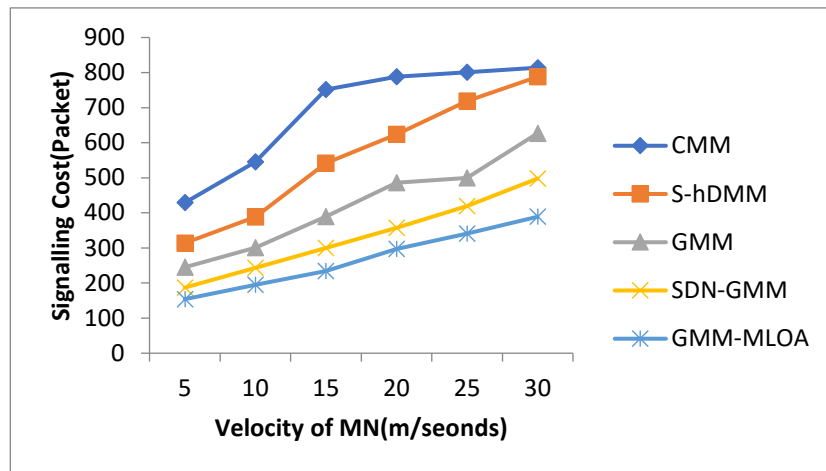


Figure 4: Handover signaling cost versus velocity of MNs

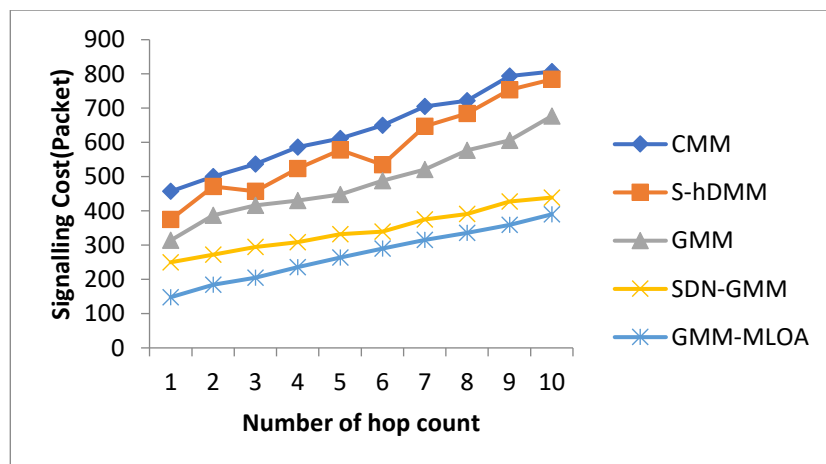


Figure 5: Handover signaling cost versus the number of hops count

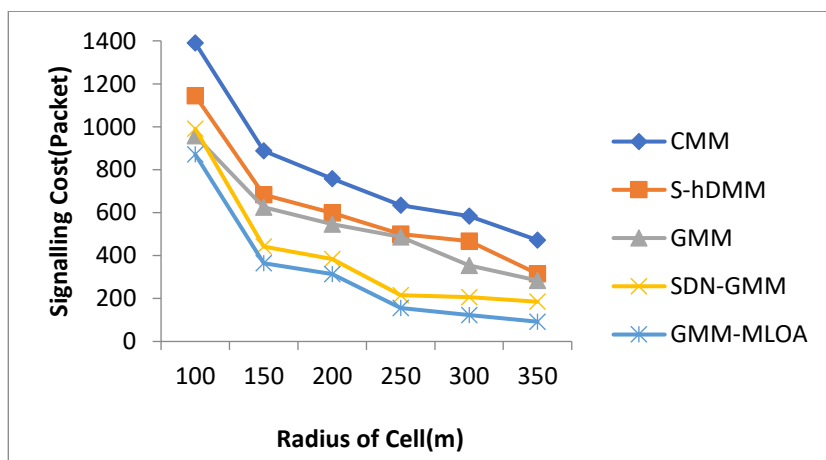


Figure 6: Handover signaling cost versus radius of cells

The result of the radius of the cell is analyzed, and it is found that as the radius gets larger, the HO occurrence rate gets lowered occurrence rate. In this analysis result, the proposed GMM-MLOA model attains an improved output of 92 signaling cost efficiency for a 350 radius of cell in metres, whereas other methods such as CMM, S-hDMM, GMM, and SDN-GMM give higher signaling costs of 472, 315, 284, and 185 for a 350 radius of cell, respectively. The comparison graph of handover signaling loss versus radius of cell using different techniques is revealed in Figure 6.

4.2 Handover latency

The HO latency is defined as the time it takes for MN to accept the first packet from its novel mobility router from the moment it begins the HO process. In order to move a packet over wired or wireless links, the delay time must be taken into account, including propagation time, transmission delay, and queueing time. A packet sent from u to v is modelled and which involves the hop count distance from the source to the destination. Assuming the wired links are steady and reliable, the packet delay time is represented by equation (26), where p is the packet size.

$$Dt_{m,n} = \text{hopcount} * \left(\frac{S_{msg}}{BW_{bnd}} + L_{pl} + W_{aq} \right) \quad (26)$$

Where S_{msg} , L_{pl} , hop count, BW_{bnd} and W_{aq} are the average message volume, latency of propagation,

Hop distance and the bandwidth accessible, and the average delay in the queue in each router. From Figure 7, the proposed GMM-MLOA model achieves 75 ms of handover latency for an average queuing delay time of 50ms, whereas other methods, such as S-hDMM, GMM, and SDN-GMM, give higher latencies of 325ms, 275ms, 185ms, and 102ms for an average queuing delay time of 50ms, respectively. The proposal SDN-GMM method exhibits lower HO latency than the traditional handover technique, which increases the average queuing delay at each router.

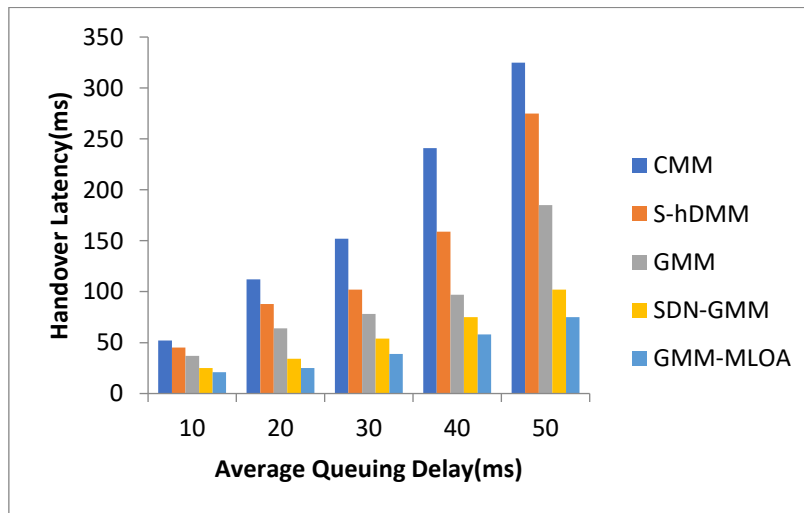


Figure 7: Handover delay versus average queuing delay

4.3 Throughput

Throughput ($thrt$) is defined as the complete volume of data which is transmitted per session ($S_{ad} * L_u$) in the particular session's delivery time, and the same is represented as in Eq(27).

$$thrt_{sdn-gmm} = S_{ad} * L_u / [(PDT_{sdn-gmm} + (S_{ad} - 1) * \rho) + T_{HO-sdn-gmm} * t_{HO-no}] \quad (27)$$

Here, S_{ad} defines the mean count of packets that are transmitted per session and L_u gives the average size of a packet from a given node to MNs, and the packet-packet time is represented as $PDT_{sdn-gmm}$ gives the time taken for the delivery of L_u number of packets. Packet delivery time (PDT) is the time it takes packets to make it from a given node to the MNs, calculated by equation (27). $T_{HO-sdn-gmm}$ is computed as a ratio in the HO μ and the mean incoming charge ϕ of the session.

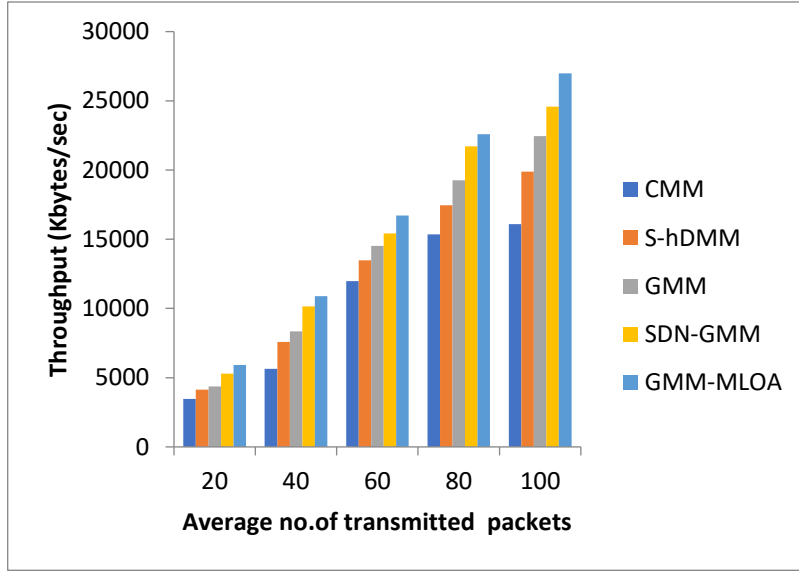


Figure 8: Throughput versus count of packets in a session

The comparison based on throughput performance of the proposed method and the conventional methods is shown in Fig. 9. Here, the packets are routed for a count of packets from 20 to 100. It shows the proposed method has a better throughput of 26987 Kbyte/sec, whereas other methods, such as S-hDMM, GMM, and SDN-GMM, give reduced throughput of 16100 Kbyte/sec, 19875 Kbyte/sec, 22457 Kbyte/sec, and 24572 Kbyte/sec for 100 transmitted packets, respectively.

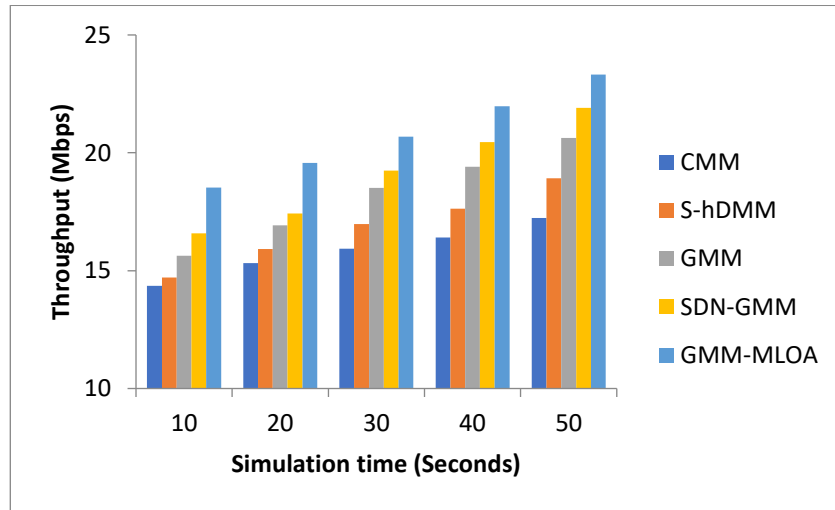


Figure 9: Network throughput comparison as a function of the simulation time

Figure 9 illustrates how network throughput and resource management performance can be compared. As can be seen from Figure 9, the network throughput trend follows the resource management extent outlined in Figure 8. Hence, the resource will be more efficiently utilized when it is managed within the network, thus resulting in a higher network throughput of 23.32 Mbps, where the conventional methods, such as S-hDMM, GMM, and SDN-GMM, have significantly lesser throughput for a given simulation time of 50 seconds.

4.4 Packet loss

In MN, during handover, the packet loss rate is proportional to either the delay in HO or the rate φ at which packets arrive per second during the HO procedure. Suppose that S_{ad} denote the mean count of the packets that are delivered in a session, then the loss rates of the proposed SDN-GMMHO are computed using Eq. (28).

$$C_{sdn-gmm} = \varphi * S_{ad} * T_{HO-delay-sdn-gmm} \quad (28)$$

Whereas $T_{HO-delay-sdn-gmm}$ denote the average HO delay of the proposed SDN-GMM method.

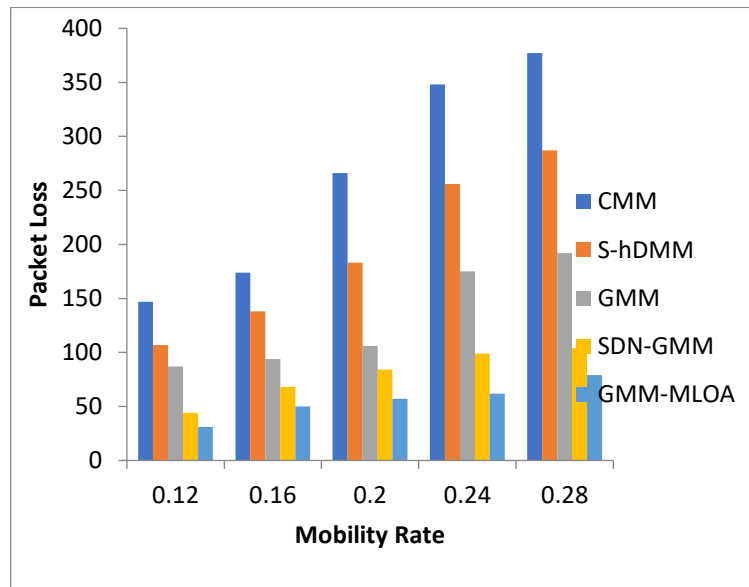


Figure 10: Packet loss versus mobility rate

According to Figure 10, the proposed method and the traditional methods result in improvements to HO rates of 0.12 to 0.28. The count of missing packets by the proposed method is partly the same as that by the existing method. The packet loss of the proposed method was 79, which is comparatively less than that of the existing methods, such as S-hDMM, GMM, and SDN-GMM, which have lesser throughput values of 377, 287 and 192, respectively, in a mobility rate of 0.28.

4.5 Transmission rate

In addition, the conventional methods show a decreased transmission rate and hence maintain a state of disconnection, whereas in the proposed method, the rate of handover is minimized, keeping the connection stable.

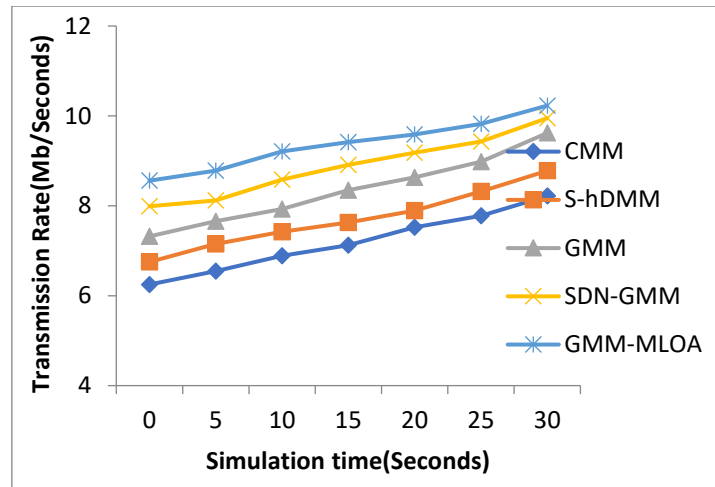


Figure 11: Transmission rate for methods

Figure 11 illustrates the transmission rate comparison of the proposed and conventional methods in terms of simulation time. Figure 11, when the HO rate is improved from 0 to 35, the transmission rate achieved by the proposed method is increased than the other methods. The proposed technique gives a higher transmission rate of 10.23 Mb/s, whereas other methods, such as S-hDMM, GMM, and SDN-GMM, give lesser throughput of 8.22 Mb/s, 8.78 Mb/s, 9.62 Mb/s, and 9.95 Mb/s against the simulation time of 30 seconds.

4.6 Delay

From Figure 12, the terminal starts to move from $t = 0$ s to 50 s. At this point in time, a handover is done where the delay is kept increasing in conventional methods and finally disconnected, whereas the proposed method found a stable state and had only a 6 ms delay in transmissions.

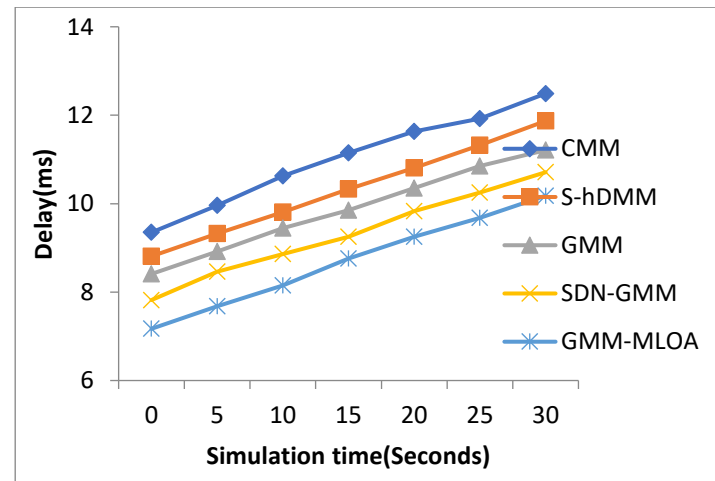


Figure 12: Delay results comparison of methods

Figure 12 illustrates the delay results comparison of the proposed method and the traditional methods in terms of simulation time. The proposed technique has a lesser delay of 10.18 ms, whereas other methods, such as S-hDMM, GMM, and SDN-GMM, give improved delays of 12.49 ms, 11.87 ms, 11.21 ms, and 10.71 ms against the simulation time of 30 seconds, respectively.

5 Conclusion and Future Work

In this study, an architecture for an SDN-enabled mobility model for seamless handovers and optimum resource management is proposed using the SDN paradigm. The Mutation Lion Optimization Algorithm (MLOA) is used for obtaining a global view of the network, which makes it more important to make decisions in an effective and optimal usage of resources. In case of seamless handovers, the MLOA are introduced for reducing the propagation of data through the early release of the OpenFlow table. In cases of optimum resource management and handovers, the proposed technique uses weights for Access Point, which are calculated based on factors like throughput, delay, and distance. Group-based Mobility Management (GMM) method, named MoMo, is proposed by utilizing Distributed Mobility Management (DMM) is then introduced to SDN. In order to prove the efficiency, the proposed method is compared with that of the conventional handover techniques over the Network Simulator 3(NS3) simulation environment. The results of the proposed method and existing methods are measured via metrics like signaling cost, handover latency, throughput, packet loss, transmission rate, and delay. In future, the proposed SDN-GMM method can be practical in the vehicular setting. The results show that SDN-based solutions represent a more effective approach for handling data traffic with high delay tolerance, as well as being a suitable enabling technology for introducing intelligence into wireless networks and providing operators with fine-grained control.

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